

# Nesterov's Smooth Minimization of Non-Smooth Functions

Michał Szewczak   Sebastian Trojan

June 23, 2026

Based on Yurii Nesterov, *Smooth minimization of non-smooth functions*, 2005.

# Motivation

Many convex optimization problems are non-smooth because their objectives contain maxima, norms, or absolute values.

Classical subgradient methods for non-smooth convex optimization have a worst-case rate of order

$$O(1/\varepsilon^2).$$

Nesterov's smoothing method uses additional max-structure of the objective. It replaces the non-smooth term by a smooth approximation and then applies an accelerated first-order method. The resulting theoretical complexity is improved to

$$O(1/\varepsilon).$$

The main trade-off is controlled by the smoothing parameter: smaller  $\mu$  gives a more accurate approximation, but a less smooth problem.

# Structured problem class

The original problem is

$$f^* = \min_{x \in Q_1} f(x),$$

where

$$f(x) = \hat{f}(x) + \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) \}.$$

- $x$  is the primal variable.
- $u$  is the auxiliary, or dual, variable.
- $\hat{f}(x)$  is the smooth part of the objective.
- $\hat{\phi}(u)$  is the part of the max-representation depending only on  $u$ .
- In the implemented examples,  $\hat{f}(x) = 0$ .

The non-smoothness comes from the maximization over  $u$ .

## Smoothing the max-term

Choose a strongly convex prox-function  $d_2$  on  $Q_2$ , normalized so that

$$\min_{u \in Q_2} d_2(u) = 0, \quad D_2 = \max_{u \in Q_2} d_2(u).$$

Define the smoothed max-term

$$f_\mu(x) = \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \}.$$

The full smoothed objective is

$$\bar{f}_\mu(x) = \hat{f}(x) + f_\mu(x).$$

The approximation error satisfies

$$\bar{f}_\mu(x) \leq f(x) \leq \bar{f}_\mu(x) + \mu D_2.$$

Therefore, choosing

$$\mu = \frac{\varepsilon}{2D_2}$$

makes the smoothing error at most  $\varepsilon/2$ .

# Gradient and smoothness trade-off

The smoothed dual response is

$$u_\mu(x) = \arg \max_{u \in Q_2} \{ \langle Ax, u \rangle - \hat{\phi}(u) - \mu d_2(u) \}.$$

Then

$$\nabla f_\mu(x) = A^* u_\mu(x), \quad \nabla \bar{f}_\mu(x) = \nabla \hat{f}(x) + A^* u_\mu(x).$$

The gradient of  $\bar{f}_\mu$  is Lipschitz continuous with constant

$$L_\mu = M + \frac{\|A\|_{1,2}^2}{\mu \sigma_2}.$$

Here  $M$  is the Lipschitz constant of  $\nabla \hat{f}$ , and  $\|A\|_{1,2}$  is the operator norm associated with the chosen primal and dual geometries.

Decreasing  $\mu$  improves approximation accuracy, but increases  $L_\mu$ .

# Accelerated minimization

For a fixed smoothing parameter  $\mu$ , the method minimizes

$$\min_{x \in Q_1} \bar{f}_\mu(x).$$

## One accelerated iteration

|           |                                                                                                         |                       |
|-----------|---------------------------------------------------------------------------------------------------------|-----------------------|
| $g_k$     | $= \nabla \bar{f}_\mu(x_k)$                                                                             | gradient,             |
| $y_k$     | $= \arg \min_{y \in Q_1} \left\{ \langle g_k, y - x_k \rangle + \frac{L_\mu}{2} \ y - x_k\ ^2 \right\}$ | local point,          |
| $G_k$     | $= \sum_{i=0}^k \alpha_i g_i, \quad \alpha_i = \frac{i+1}{2}$                                           | accumulated gradient, |
| $z_k$     | $= \arg \min_{x \in Q_1} \left\{ \frac{L_\mu}{\sigma_1} d_1(x) + \langle G_k, x \rangle \right\}$       | aggregate point,      |
| $x_{k+1}$ | $= \frac{2}{k+3} z_k + \frac{k+1}{k+3} y_k$                                                             | next search point.    |

$y_k$  uses the current gradient, while  $z_k$  uses all gradients accumulated in  $G_k$ .

# Primal-dual certificate

The algorithm builds an averaged dual point

$$\hat{u} = \frac{\sum_{k=0}^N \alpha_k u_k}{\sum_{k=0}^N \alpha_k}, \quad u_k = u_\mu(x_k), \quad \hat{x} = x_N.$$

The dual lower-bound function is

$$\phi(u) = \min_{x \in Q_1} \{\hat{f}(x) + \langle Ax, u \rangle\} - \hat{\phi}(u).$$

By weak duality,

$$\phi(\hat{u}) \leq f^* \leq f(\hat{x}).$$

Therefore the primal-dual gap

$$f(\hat{x}) - \phi(\hat{u})$$

certifies the solution quality.

# Theoretical iteration bound

Nesterov's estimate gives

$$f(\hat{x}) - \phi(\hat{u}) \leq \frac{4\|A\|_{1,2}}{N+1} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}} + \frac{4MD_1}{\sigma_1(N+1)^2}.$$

In the implemented examples,  $\hat{f} = 0$ , so  $M = 0$ . Hence

$$f(\hat{x}) - \phi(\hat{u}) \leq \frac{4\|A\|_{1,2}}{N+1} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}}.$$

To guarantee  $f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon$ , it is sufficient to take

$$N_{\text{pred}} + 1 = \left\lceil \frac{4\|A\|_{1,2}}{\varepsilon} \sqrt{\frac{D_1 D_2}{\sigma_1 \sigma_2}} \right\rceil.$$

Thus the predicted complexity is

$$N = O(1/\varepsilon).$$

**Matrix game:**  $\min_{x \in \Delta_n} \max_{u \in \Delta_m} u^\top A x$

**Continuous location:**  $\min_{\|x\|_2 \leq r} \sum_{j=1}^p w_j \|x - c_j\|_2$

**Maximum of absolute values:**  $\min_{\|x\|_2 \leq r} \max_{1 \leq j \leq m} |a_j^\top x - b_j|$

**Sum of absolute values:**  $\min_{\|x\|_2 \leq r} \sum_{j=1}^m |a_j^\top x - b_j|$

# Experimental plan

- 1 Reproduce Nesterov's matrix game experiment.
- 2 Apply the same scheme to the other three examples.
- 3 Test the empirical dependence on  $\varepsilon$ .
- 4 Compare standard smoothing with a changing- $\mu$  strategy.

In every experiment, convergence is measured by the certified condition

$$f(\hat{x}) - \phi(\hat{u}) \leq \varepsilon.$$

# Experiment 1: matrix game reproduction

Reproduced problem:

$$\min_{x \in \Delta_n} \max_{u \in \Delta_m} u^\top A x, \quad A_{ij} \sim U[-1, 1].$$

Tested grid:

- $\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ ,
- $m \in \{100, 300, 1000\}$ ,
- $n \in \{100, 300, 1000, 3000, 10000\}$ , with the largest  $n$  omitted for  $\varepsilon = 10^{-4}$ .

Each table cell reports iterations / percentage of the predicted iteration bound.

# Experiment 1: matrix game results

| $\varepsilon = 10^{-2}$ |            |            |            |             |            |
|-------------------------|------------|------------|------------|-------------|------------|
| $m \backslash n$        | 100        | 300        | 1 000      | 3 000       | 10 000     |
| 100                     | 765/42%    | 934/46%    | 1 108/49%  | 1 187/49%   | 1 281/49%  |
| 300                     | 690/34%    | 910/40%    | 1 126/45%  | 1 268/47%   | 1 475/51%  |
| 1 000                   | 761/34%    | 873/35%    | 1 086/39%  | 1 284/43%   | 1 457/46%  |
| $\varepsilon = 10^{-3}$ |            |            |            |             |            |
| $m \backslash n$        | 100        | 300        | 1 000      | 3 000       | 10 000     |
| 100                     | 7 123/39%  | 8 426/41%  | 8 928/40%  | 10 191/42%  | 10 080/39% |
| 300                     | 7 337/36%  | 8 639/38%  | 10 154/40% | 11 302/42%  | 12 022/41% |
| 1 000                   | 7 526/33%  | 8 828/35%  | 10 157/37% | 11 595/39%  | 14 167/44% |
| $\varepsilon = 10^{-4}$ |            |            |            |             |            |
| $m \backslash n$        | 100        | 300        | 1 000      | 3 000       |            |
| 100                     | 68 505/37% | 64 512/31% | 77 136/34% | 74 024/30%  |            |
| 300                     | 60 932/30% | 78 680/34% | 83 844/33% | 92 862/34%  |            |
| 1 000                   | 70 778/31% | 77 644/31% | 93 346/34% | 102 925/35% |            |

## Experiment 2: other problem classes

The matrix game experiment was extended to:

- Continuous location:  
 $\min_{\|x\|_2 \leq r} \sum_{j=1}^p w_j \|x - c_j\|_2, \quad c_j \sim \text{Unif}(B_2(0, 0.8r)), \quad w_j \sim U[0.5, 1.5], \quad r = 1.$   
Grid:  $p \in \{10, 50\}, d \in \{2, 5\}$  where  $c_j, x \in \mathbb{R}^d$ .
- Maximum of absolute values:  $\min_{\|x\|_2 \leq r} \max_{1 \leq j \leq m} |a_j^\top x - b_j|, \quad A_{ji}, b_j \sim U[-1, 1], \quad r = 1.$   
Grid:  $m \in \{100, 300\}, n \in \{50, 200\}$  where  $a_j, x \in \mathbb{R}^n$ .
- Sum of absolute values:  $\min_{\|x\|_2 \leq r} \sum_{j=1}^m |a_j^\top x - b_j|, \quad A_{ji}, b_j \sim U[-1, 1], \quad r = 1.$   
Grid:  $m \in \{100, 300\}, n \in \{50, 200\}$  where  $a_j, x \in \mathbb{R}^n$ .

Common accuracy levels:

$$\varepsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}.$$

## Experiment 2: summary tables

| Continuous location        |            |           |        |       |
|----------------------------|------------|-----------|--------|-------|
| $\epsilon$                 | Avg. iter. | Max iter. | Avg. % | Time  |
| $10^{-4}$                  | 171 599    | 320 867   | 30.64% | 31.8s |
| $10^{-3}$                  | 20 525     | 31 653    | 43.74% | 3.86s |
| $10^{-2}$                  | 1 988      | 3 606     | 36.70% | 0.35s |
| Maximum of absolute values |            |           |        |       |
| $\epsilon$                 | Avg. iter. | Max iter. | Avg. % | Time  |
| $10^{-4}$                  | 171 486    | 249 861   | 39.81% | 10.6s |
| $10^{-3}$                  | 20 032     | 33 671    | 44.76% | 1.26s |
| $10^{-2}$                  | 2 593      | 3 788     | 56.06% | 0.16s |
| Sum of absolute values     |            |           |        |       |
| $\epsilon$                 | Avg. iter. | Max iter. | Avg. % | Time  |
| $10^{-4}$                  | 1 198 404  | 2 711 368 | 29.45% | 54.9s |
| $10^{-3}$                  | 135 894    | 284 262   | 32.52% | 6.43s |
| $10^{-2}$                  | 12 088     | 27 750    | 30.43% | 0.51s |

## Experiment 3: testing the $O(1/\varepsilon)$ behavior

To isolate the dependence on  $\varepsilon$ , five seeds were used; for each seed, one instance was generated and then kept fixed across all values of  $\varepsilon$ :

$$\varepsilon \in \{10^{-1}, 5 \cdot 10^{-2}, 10^{-2}, 5 \cdot 10^{-3}, 10^{-3}, 5 \cdot 10^{-4}, 10^{-4}\}.$$

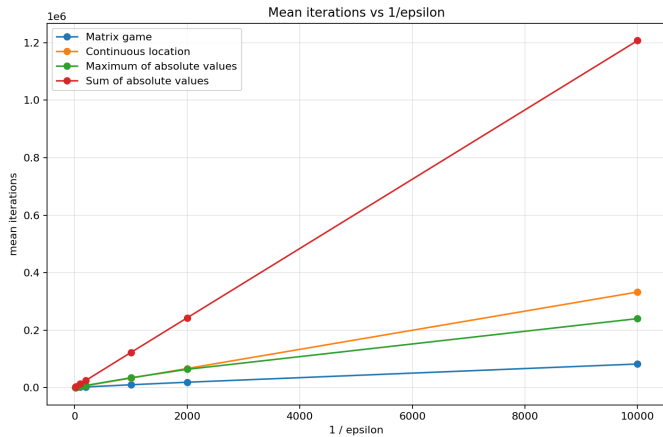
The theory predicts

$$N(\varepsilon) \approx C \frac{1}{\varepsilon}.$$

Therefore:

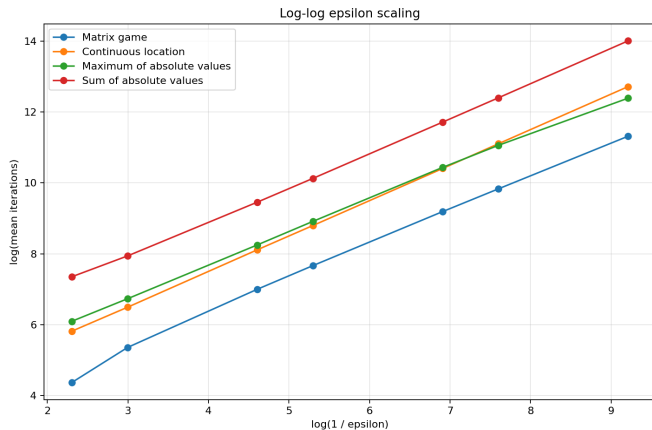
- iterations versus  $1/\varepsilon$  should be approximately linear,
- the log-log fitted exponent  $\hat{p}$  should be close to 1.

## Experiment 3: iterations versus $1/\varepsilon$



The curves are close to linear; the difference between problems is mainly the constant factor.

## Experiment 3: log-log scaling



Empirical power-law model:

$$N(\epsilon) \approx C\epsilon^{-p}.$$

Taking logarithms gives

$$\log N = \log C + p \log(1/\epsilon).$$

Therefore  $\hat{p}$  is the slope of the fitted line on the log-log plot. The theory predicts

$$p = 1.$$

| Problem             | $\hat{p}$ | SE     |
|---------------------|-----------|--------|
| Matrix game         | 0.9897    | 0.0096 |
| Continuous location | 0.9986    | 0.0054 |
| Max. abs. values    | 0.9208    | 0.0042 |
| Sum abs. values     | 0.9649    | 0.0096 |

# Experiment 4: changing the smoothing parameter

## 1. Motivation

- Standard smoothing uses  $\mu_\varepsilon = \varepsilon/(2D_2)$  from the beginning.
- For small  $\varepsilon$ , this value is small, so the smoothed problem is sharper and harder.

## 2. Method

- Start with a larger smoothing parameter and decrease it geometrically:

$$\mu_0 = c\mu_\varepsilon, \quad \mu_{t+1} = \rho\mu_t, \quad c > 1, \quad \rho \in (0, 1).$$

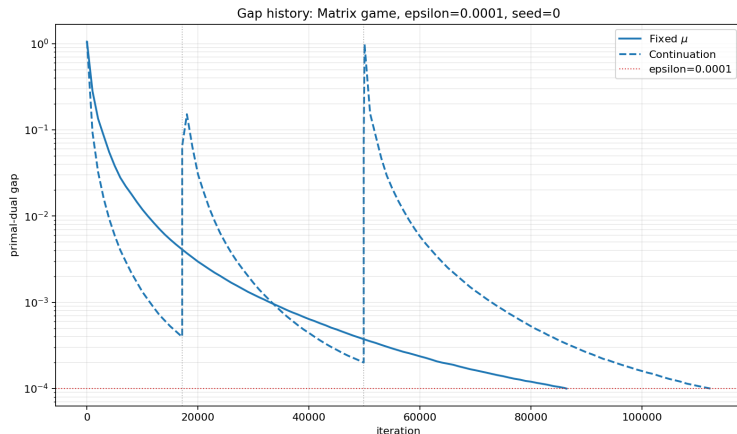
- Move to the next stage when

$$f(\hat{x}) - \phi(\hat{u}) \leq c_{\text{stage}}\mu_t D_2, \quad c_{\text{stage}} \geq 1.$$

## 3. Our parameters

$$c = 8, \quad \rho = 0.5, \quad c_{\text{stage}} = 1, \quad 8\mu_\varepsilon \rightarrow 4\mu_\varepsilon \rightarrow 2\mu_\varepsilon \rightarrow \mu_\varepsilon.$$

## Experiment 4: gap history



After reducing  $\mu$ , the smoothed objective changes and the primal-dual gap can jump upward. In most tested cases, continuation required more iterations.

- The implementation reached the required primal-dual gap on all reported instances.
- Matrix game results reproduce the qualitative behavior from Nesterov's paper.
- The empirical scaling is consistent with the theoretical  $O(1/\varepsilon)$  complexity.
- Worst-case iteration bounds were conservative on random instances.
- Practical difficulty depends strongly on the problem structure.
- The tested changing- $\mu$  strategy was usually less efficient than the standard smoothing method.

-  [Yurii Nesterov.](#)  
*Smooth minimization of non-smooth functions.*  
[Mathematical Programming](#), 103(1):127–152, 2005.